

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Medium

Question

Distance (kilometers)	Average time (minutes)
0.32	8
0.56	14
0.68	17

The table gives the average time t , in minutes, it takes Carly to travel a certain distance d , in kilometers. Which equation could represent this linear relationship?

Answer

- A. $t = 4d$
- B. $t = \frac{1}{25}d$
- C. $t = 25d$
- D. $t = \frac{1}{4}d$

Correct Answer: C

Rationale

Choice C is correct. The average time t , in minutes, it takes Carly to travel a certain distance d , in kilometers, is given in the table. This linear relationship can be represented by an equation in the form $t = ad + b$, where a and b are constants. The table shows that it takes Carly an average time of 8 minutes to travel 0.32 kilometers. Substituting 8 for t and 0.32 for d in the equation $t = ad + b$ yields $8 = 0.32a + b$. Subtracting $0.32a$ from both sides of this equation yields $8 - 0.32a = b$. The table also shows that it takes Carly an average time of 14 minutes to travel 0.56 kilometers. Substituting 14 for t and 0.56 for d in the equation $t = ad + b$ yields $14 = 0.56a + b$. Subtracting $0.56a$ from both sides of this equation yields $14 - 0.56a = b$. Substituting $8 - 0.32a$ for b in this equation yields $14 - 0.56a = 8 - 0.32a$. Subtracting 8 from both sides of this equation yields $6 - 0.56a = -0.32a$. Adding $0.56a$ to both sides of this equation yields $6 = 0.24a$. Dividing both sides of this equation by 0.24 yields $25 = a$. Substituting 25 for a in the equation $8 = 0.32a + b$ yields $8 = 0.32(25) + b$, or $8 = 8 + b$. Subtracting 8 from both sides of this equation yields $0 = b$. Substituting 25 for a and 0 for b in the equation $t = ad + b$ yields $t = 25d + 0$, or $t = 25d$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.